Teal PDU Replacements

- **sys** none **propclass**

1 Personnel Requirements

Required Persons: 1
Preliminary Timing (mins): **preltime**
Procedure Timing (mins): Multiple (See Procedure Overview)
Finalization Timing (mins): **finaltime**

2 Procedure Overview

This document covers replacing the following FRU's for the XGD Power Distribution Unit (PDU):

- 420V E-Stop Contactor (CON1 and CON2)—Timing: 40 minutes
- 48V Power Supplies—Timing: 30 minutes
- 48V Fan with Tach (FAN1 and FAN2)—Timing: 30 minutes

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item: Service Tool Kit

Part#: 5112581

Qty: 1

Item: Extra Long Slotted Screwdriver

Qty: 1

Item: Extra Long Phillips Screwdriver

Qty: 1

3.2 Consumables

Item: Tie Wraps

Qty: AR

3.3 Replacement Parts

Item: 420V E-Stop Contactor (CON1 and CON2)

Part#: 5140621-3

Qty: 1

Item: 48V Power Supplies

Part#: 5140621-4

Qty: 1

Item: 48V Fan with Tach (FAN1 and FAN2)

Part#: 5140621-5

Qty: 1

Item: Power Supply Unit

Part#: 5140621-4

Qty: 1

Item: Control PC Board Asm

Part#: 5140621-2

Qty: 1

3.4 Safety

DANGER

ELECTROCUTION HAZARD!
HIGH VOLTAGE PRESENT.
USE PROPER LOCK OUT / TAG OUT PROCEDURES BEFORE SERVICING AND/OR MAINTAINING.

3.5 Required Conditions

No required conditions

4 Procedure

4.1 Removing the Front Panel

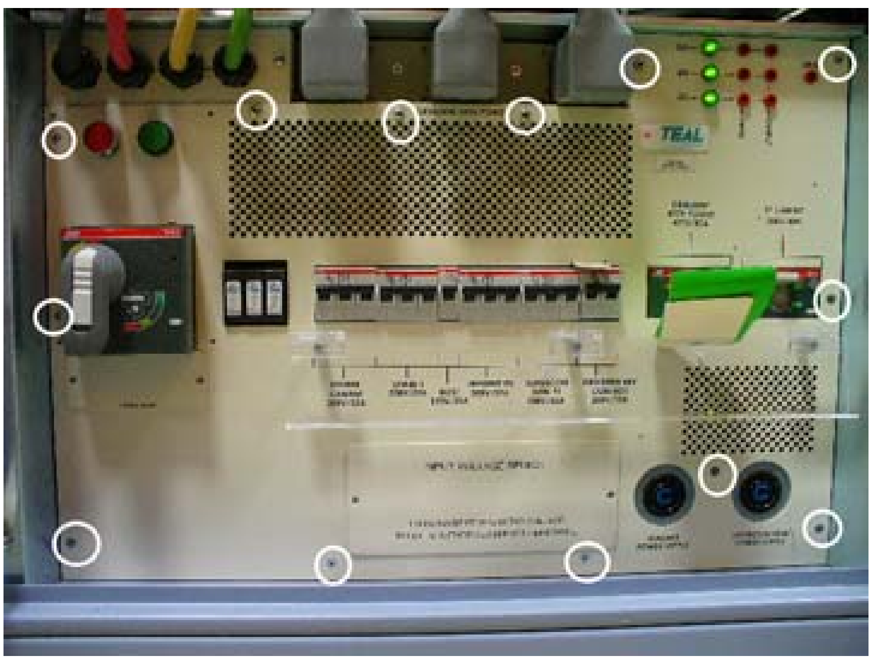
1. Perform LOTO for the PDU. For more information, see the *LOTO for the PGR PDU/Gradient Subsystem* document.

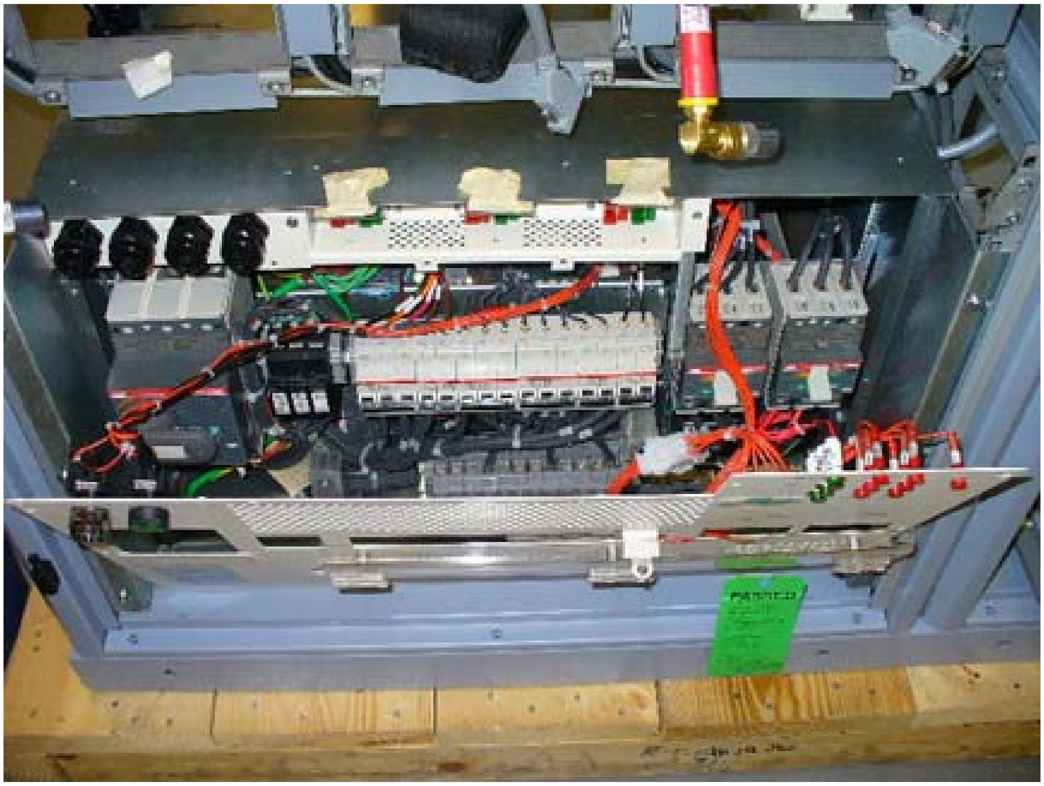
2. Remove the four screws on the E-Stop panel. See illustration below.

Illustration 1: E-Stop Panel Removal

3. Remove the 13 screws to remove the front panel of the PDU. See illustration below.

Illustration 2: Front Panel Removal





4. The panel is still connected to the PDU. Ensure you unlatch it to remove the panel completely. See illustration above.

4.2 Replacing the 420V E-Stop Contactor (CON1 or CON2)

The CON1 and CON2 contactors are stacked one on top of the other held in place by a single vertical rail at the rear of each. Located just underneath the Teal TVSS modules, it is easier to replace CON1 than CON2 because CON1 is located on top. However, replacing CON2 requires that you remove CON1 first since CON2 is on the bottom.

1. Perform LOTO for the PDU. For more information, see the *LOTO for the PGR PDU/Gradient Subsystem* document.
2. Using a Phillips screwdriver, remove the PDU front panel.
3. Using an extra long Phillips screwdriver, loosen the set screws for each contact pole located on the left, top, and right sides of CON1. **Note**: When working with CON2, there are contact poles on the left and right sides only.

Illustration 3: CON1 Contact Pole Locations, Left Side

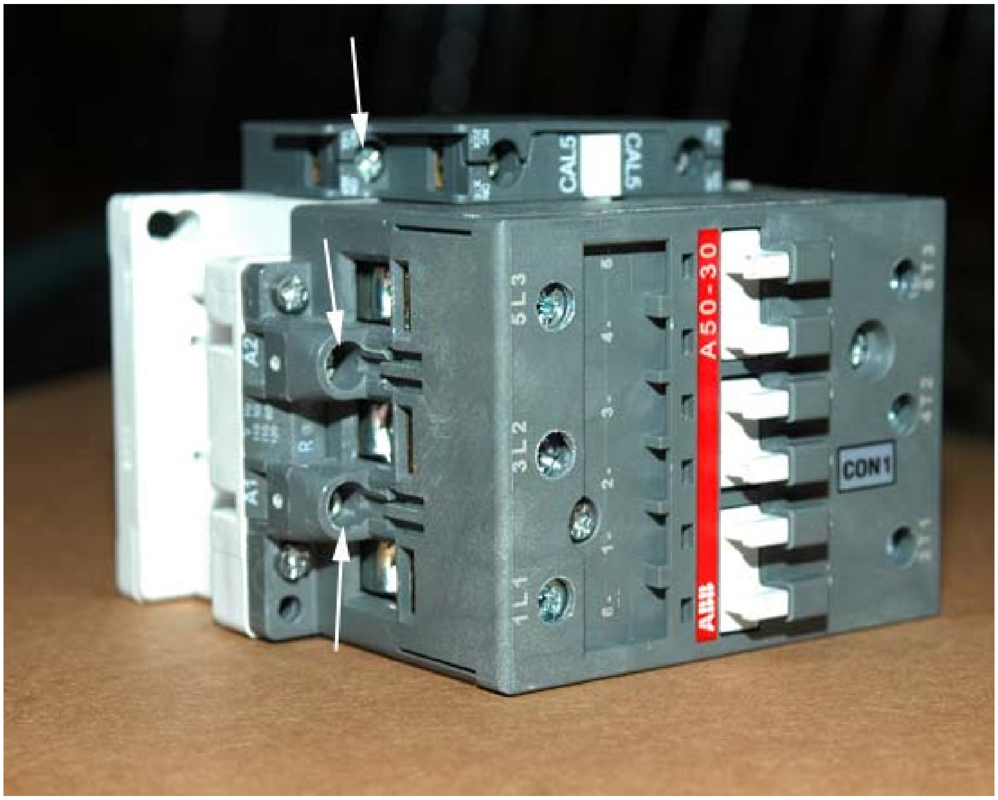
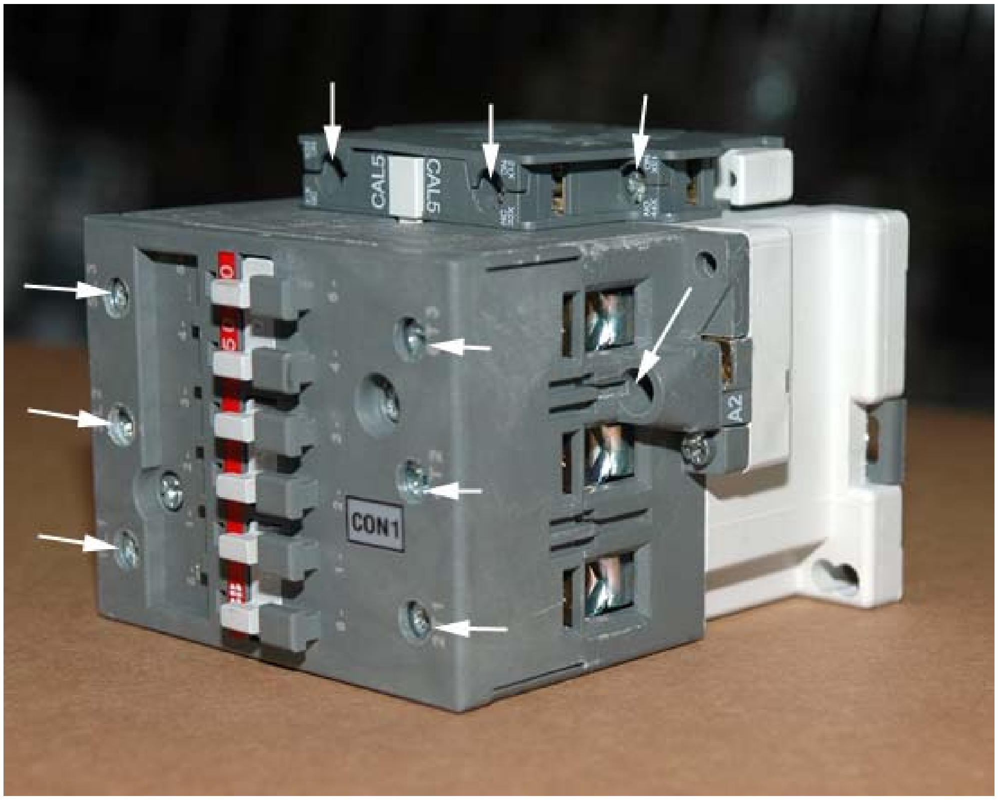


Illustration 4: CON1 Contact Pole Locations, Right Side



4. Using an extra long slotted screwdriver, place the blade of the screwdriver in the vertical slot (latch) on the right side of CON1 or CON2, and slide the latch right to release the contactor from the rail.

Illustration 5: Slide Latch, Right Side of CON1

5. Position the new contactor (CON1 or CON2) on the rail by angling the front right side out so that the rear left guide of the contactor connects to the rail first, then slowly push in the front right side until the latch snaps, thereby indicating that the rear right guide is secure on the rail.

6. Replace each wire into their respective contact pole and then tighten each set screw.

7. Replace the PDU front panel.

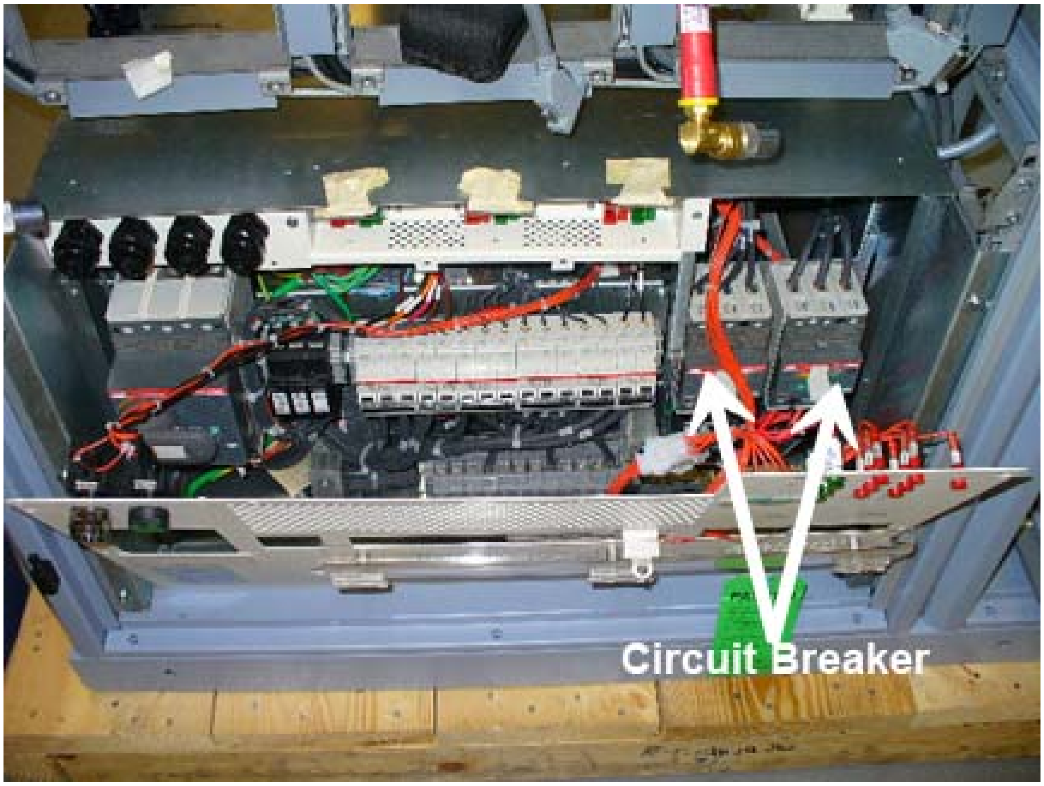
8. Remove LOTO for the PDU. Refer to the *LOTO for the PGR PDU/Gradient Subsystem* document.

4.3 Replacing the 48V Power Supply

1. Perform LOTO for the PDU. For more information, see the *Perform LOTO for the PDU. For more information, see the*

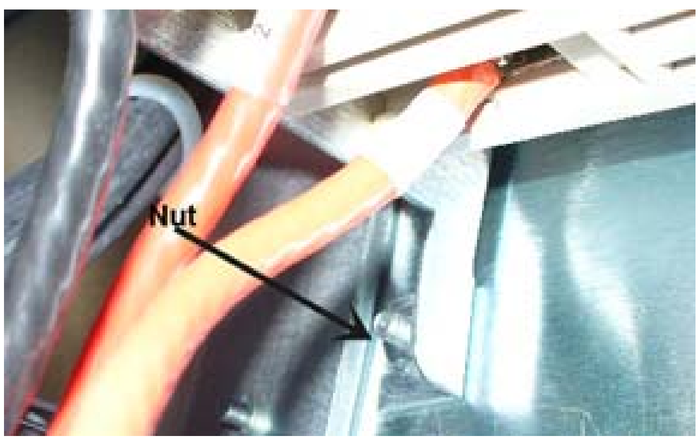
2. The Power Supply Unit is behind the Circuit Breaker. See illustration below.

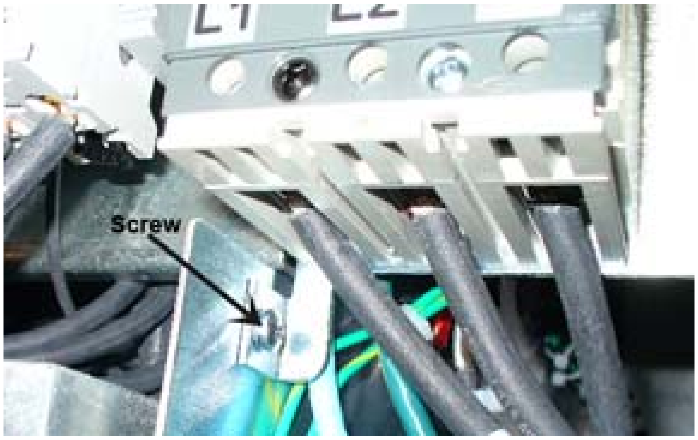
Illustration 6: Position of Power Supply Unit



3. Remove the two (2) screws and two (2) nuts, one on either side, to remove the Circuit Breaker. This is in front of the power supply unit in the PDU.

Illustration 7: Removing the Circuit Breaker

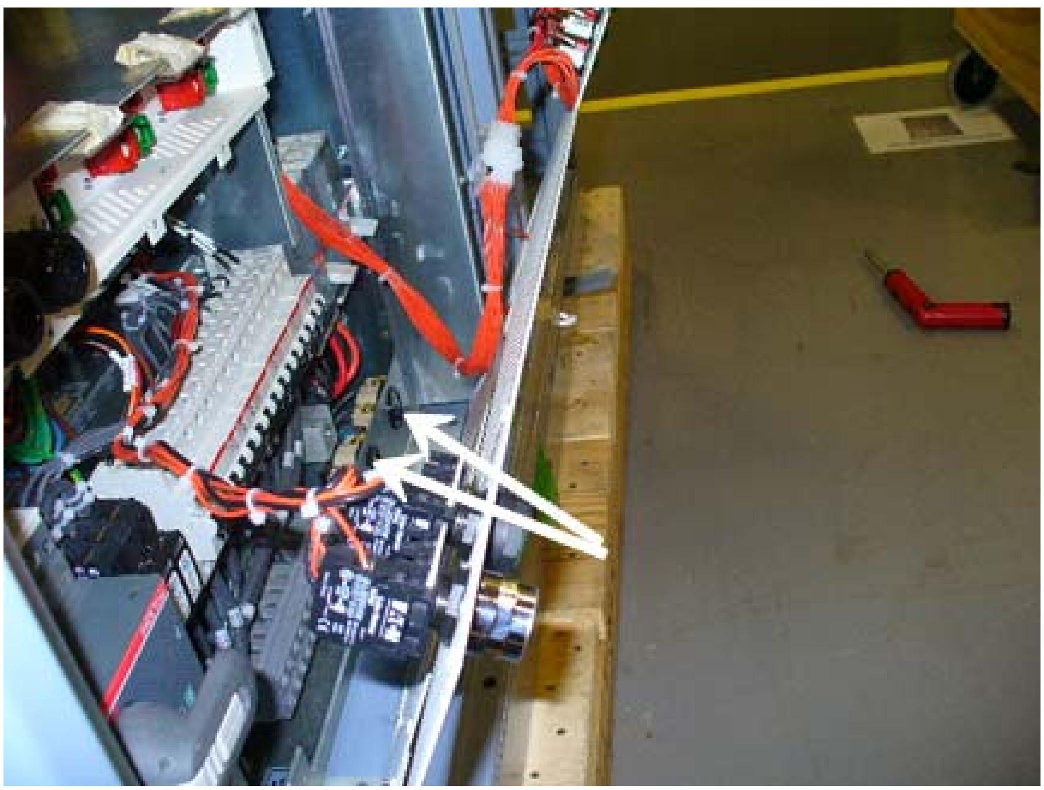




4. The Circuit Breaker is still connected to the PDU with wires. Move the Circuit Breaker out of the way to remove the Power Supply Unit out of the PDU.

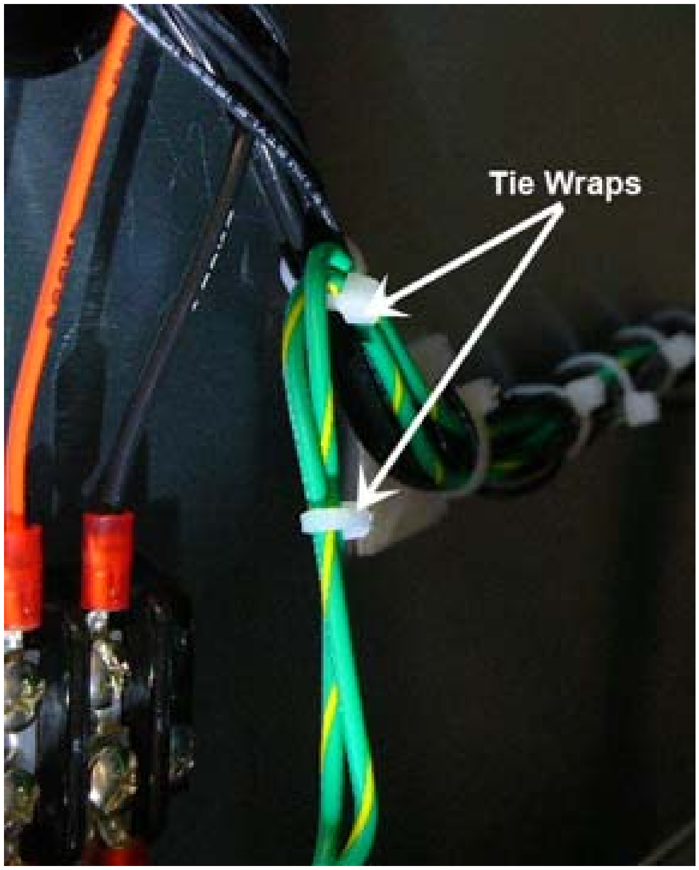
5. Remove two (2) screws and remove the panel for the magnet power supply and the super con shim power supply. See Illustrations below. Then remove the two screws from the Power Supply Unit panel.

Illustration 8: Removing the Power Supply Unit



6. Cut the tie wraps, so the power supply unit can be pulled forward.

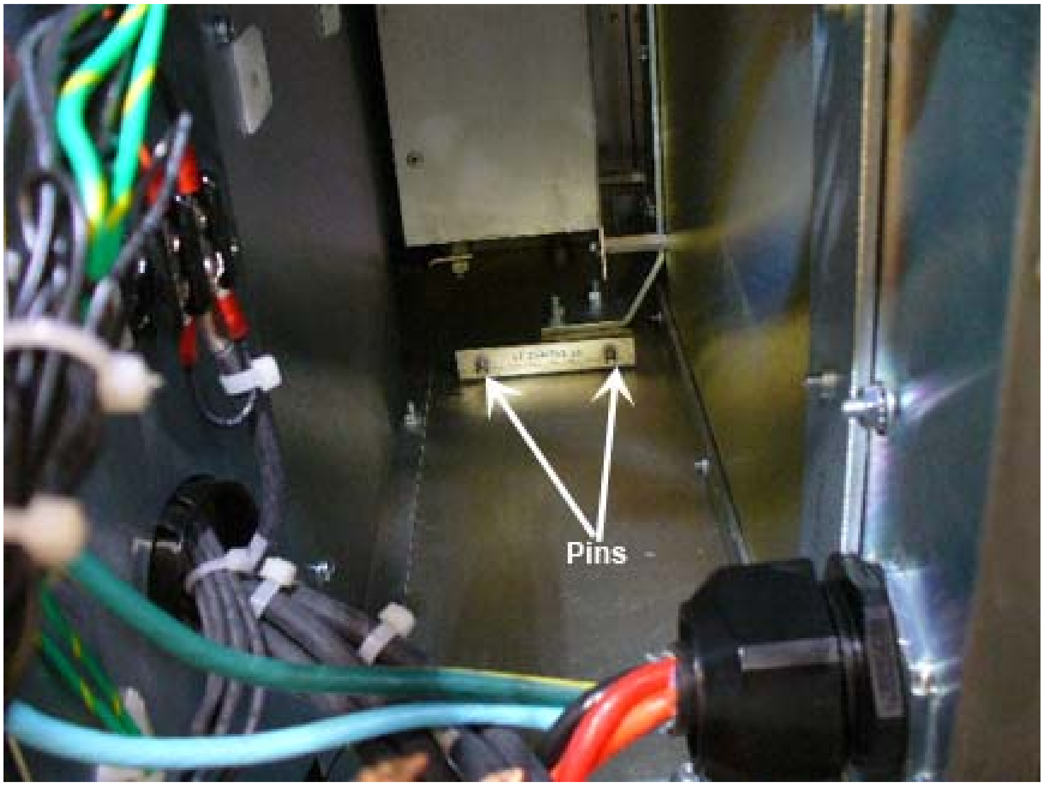
Illustration 9: Tie Wrap



7. Lift the power supply and then take it out of the PDU cabinet. Note down the connection to the power supply unit before disconnecting them. See

8. To put the Power Supply Unit back into the PDU, ensure that the power supply unit slots fit into the pins. See illustration below.

Illustration 10: Pins in the PDU



9. Tighten the screws and put the tie wraps back in place.

10. Put the Circuit Breaker back in place and tighten the screws.

11. Remove LOTO for the PDU. Refer to the *Remove LOTO for the PDU. Refer to the*

4.4 Replacing the 48V Fan with Tach (FAN1 and FAN2)

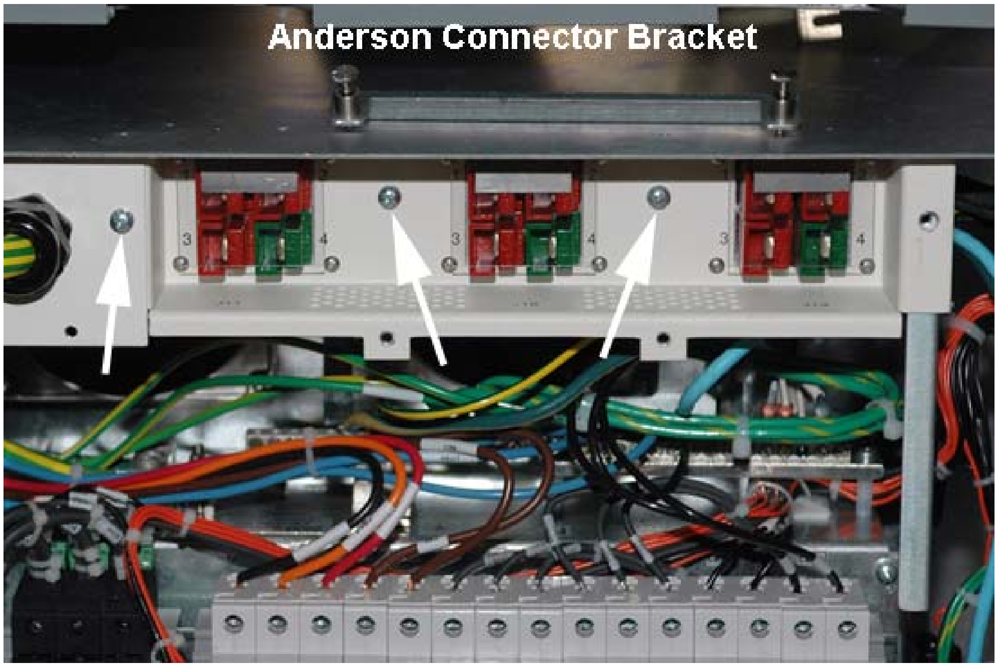
The fans (FAN1 and FAN2) are daisy-chained so if one fails, the FRU is set up to replace both fans. The fans are located behind the Anderson connector bracket.

1. Perform LOTO for the PDU. For more information, see the *LOTO for the PGR PDU/Gradient Subsystem* document.

2. Using a Phillips screwdriver, remove the PDU front panel.

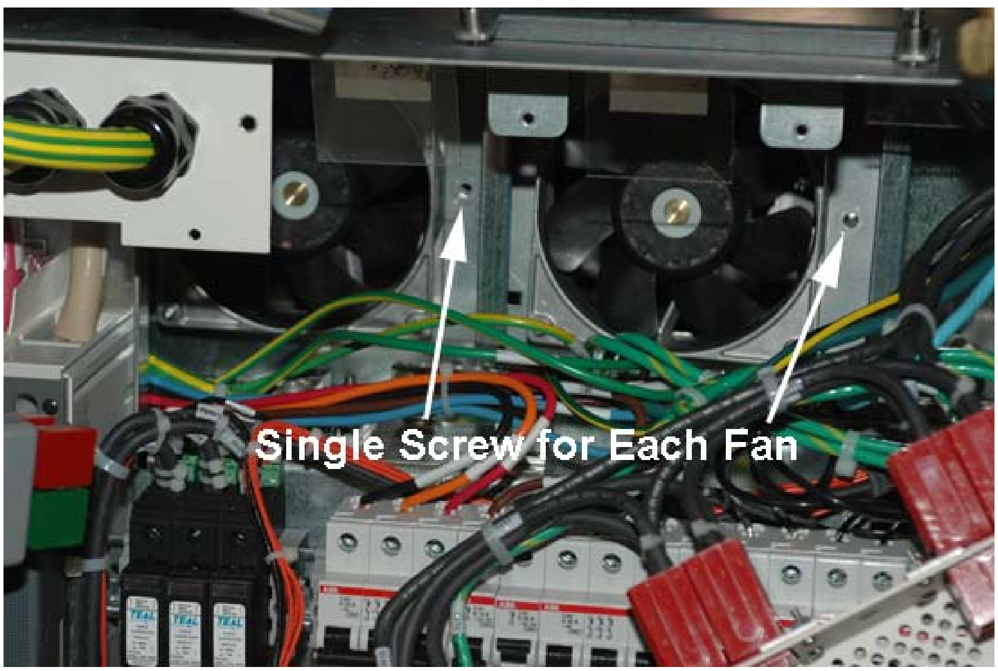
3. Using a Phillips screwdriver, remove three screws holding the Anderson connector bracket in place, remove it and set it aside.

Illustration 11: Anderson Connector Bracket



4. Using a Phillips screwdriver, remove the screw holding the fan bracket assembly in place for each fan.

Illustration 12: FAN1 and FAN2 Screw Location



5. Disconnect the fan's wire connection (power and sensor) from the harness. Remove the fans.

6. Position the new fans in place and replace each bracket screw. Reconnect the fan's wire connection in the harness.

7. Reposition the Anderson connector bracket and replace the screws.

8. Replace the PDU front panel.

9. Remove LOTO for the PDU. Refer to the *LOTO for the PGR PDU/Gradient Subsystem* document.

5 Finalization

No finalization steps